

SU(3) color

$\Delta^{++} = u \uparrow u \uparrow u \uparrow \Rightarrow$ introduce color quantum number.

3 colors: R G B

Hypothesis: physical states are colorless or color singlet

$$\Psi_{\Delta^{++}} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{flavor}} \psi_{\text{color}}$$

$L=0$ $S=1/2$ S A

$$\psi_{\text{color}} = \frac{1}{\sqrt{6}} [RGB + BRG + GBR - BGR - GRB - RGB]$$

color singlet

SU(3) color group $\Rightarrow$

8 generators
gluons: mediators of QCD
massless, $S=1$

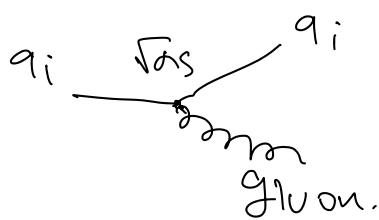
Yang-Mills theory

features of color: $R+G+B=0$

quarks: fundam. repr. 3 of SU(3) color

anti-quarks: fundam. conjug. repr. of SU(3) $\bar{3}$ $\bar{R}, \bar{G}, \bar{B}$

$$R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad G = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



$$\alpha_s = \frac{g_s^2}{4\pi}$$

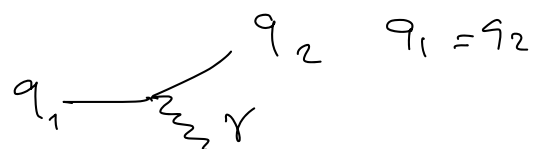
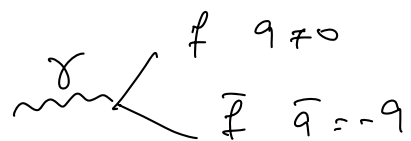
α_s : QCD/strong coupling constant

g_s : color charge.

8 gluons: carry color charge

QED: γ has no electric charge

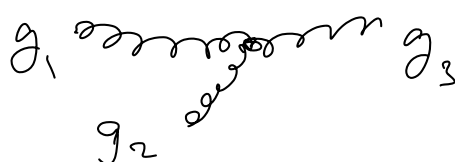
QED



QCD

total color is conserved (SU(3) group symm)

gluons can interact



Gluons

color of gluons from $3 \times \bar{3} = \cancel{1} + 8$
 $\xrightarrow{\text{singlet color}} \text{color octet}$

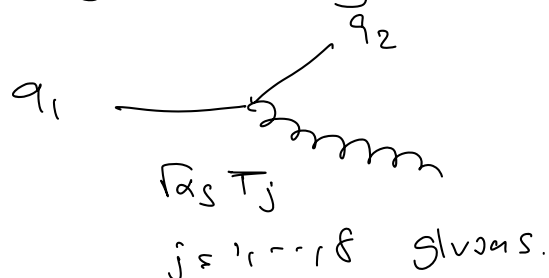
there is no colorless gluon. $\Rightarrow$ free colorless gluons around if they existed

color singlet: $\frac{1}{\sqrt{3}}(R\bar{R} + B\bar{B} + G\bar{G})$ this does not exist

gluon octet: $R\bar{G}, R\bar{B}, G\bar{R}, G\bar{B}, B\bar{R}, B\bar{G},$

$$\frac{1}{\sqrt{2}}(R\bar{R} - G\bar{G}), \frac{1}{\sqrt{6}}(R\bar{R} + G\bar{G} - 2B\bar{B})$$

gluons carry both color and anti-color.



T_j : generators in $SU(3)_{\text{color}}$.

$$R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad G = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Carter + Weyl, in early 1900 matrices for T_j

Gell-Mann matrices

$$T_j = \frac{1}{2} \lambda_j$$

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_3 \sim R\bar{R} - G\bar{G}$$

$$\lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}; \quad \lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}; \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

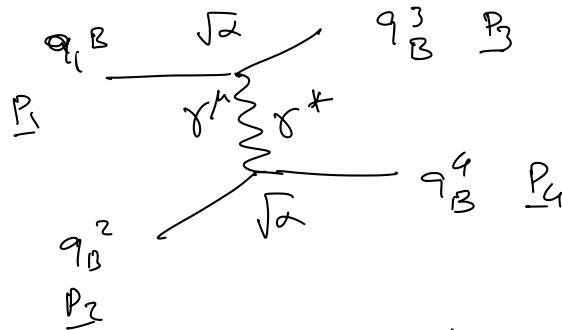
$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}; \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$\lambda_8 = R\bar{R} + G\bar{G} - 2B\bar{B}$$

λ_1 :

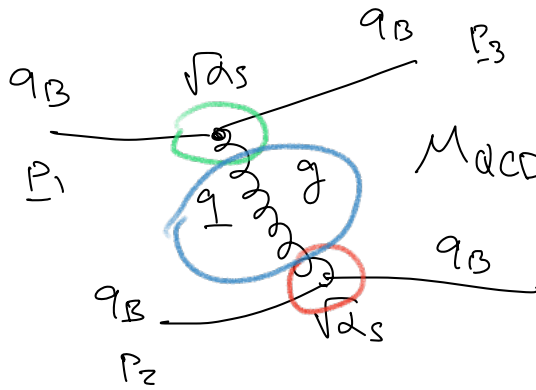


$$q_B^1 + q_B^2 \rightarrow q_B^3 + q_B^4$$



$$\alpha = \frac{e^2}{4\pi}$$

$$M_{QED} \sim (\sqrt{\alpha})^2 \bar{u}(p_3) \gamma^\mu u(p_1) \underbrace{\Pi(q^2)}_{\gamma \text{ propagator}} \bar{u}(p_4) \gamma^\nu u(p_2)$$



$$M_{QCD} \sim (\sqrt{\alpha_s})^2 \bar{u}(p_3) \gamma^\mu \frac{\lambda_8}{2} u(p_1) \Pi(q^2) \bar{u}(p_4) \gamma^\nu \frac{\lambda_8}{2} u(p_2)$$

g propagator

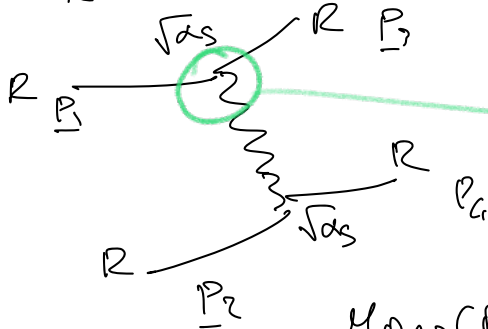
$$QCD \quad (\sqrt{\alpha_s})^2 \frac{1}{2} \overline{001} \begin{pmatrix} \lambda_j \\ \lambda_8 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2} \left(-\frac{2}{\sqrt{3}} \right) (\sqrt{\alpha_s})^2$$

$$= -\frac{1}{\sqrt{3}} \alpha_s$$

$$M_{QCD}(BB \rightarrow BB) \sim \left(-\frac{1}{\sqrt{3}} \sqrt{\alpha_s} \right)^2 = \frac{1}{3} \alpha_s$$

color factor for 1 vertex

$$q_R + q_R \rightarrow q_R + q_R \quad R = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$



$$\overline{100} (\lambda_3 + \lambda_8) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{\sqrt{\alpha_s}}{2}$$

$$M_{QCD}(RR \rightarrow RR) = M_3 + M_8$$

$$M_3 \sim \frac{\sqrt{\alpha_s}}{2} \overline{100} \lambda_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{\sqrt{\alpha_s}}{2} \overline{100} \lambda_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \left(\frac{\sqrt{\alpha_s}}{2} \right)^2 (1)^2$$

$$M_8 \sim \frac{\sqrt{\alpha_s}}{2} \overline{100} \lambda_8 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \frac{\sqrt{\alpha_s}}{2} \overline{100} \lambda_8 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \left(\frac{\sqrt{\alpha_s}}{2} \right)^2 \left(\frac{1}{\sqrt{3}} \right)^2$$

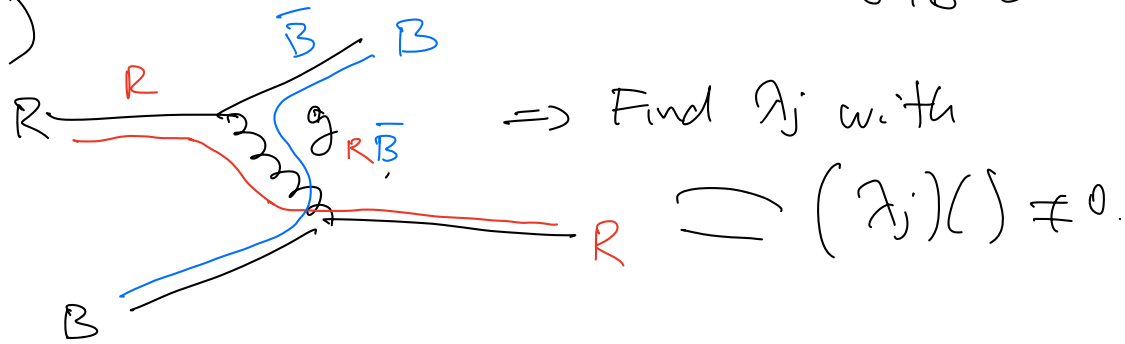
$$M_{QCD}(RR \rightarrow RR) \sim \frac{\alpha_s}{4} \left(1 + \frac{1}{3} \right) = \frac{\alpha_s}{4} \frac{4}{3} = \frac{\alpha_s}{3}$$

$$M(R\bar{R} \rightarrow R\bar{R}) \sim$$

$$\frac{\alpha_s}{4} \left[1 + \frac{1}{3} \right]$$

$$\begin{aligned} R+B+G &= 0 \\ R+\bar{R} &= 0 \\ B+\bar{B} &= 0 \\ G+\bar{G} &= 0 \end{aligned}$$

$$M(RB \rightarrow ?)$$



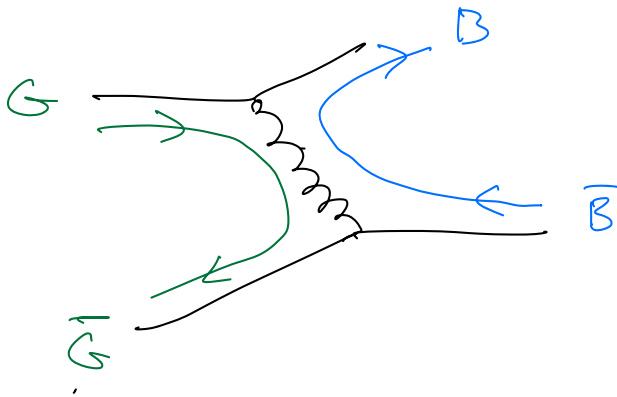
Matrix Element

$$LO \propto \alpha_s$$

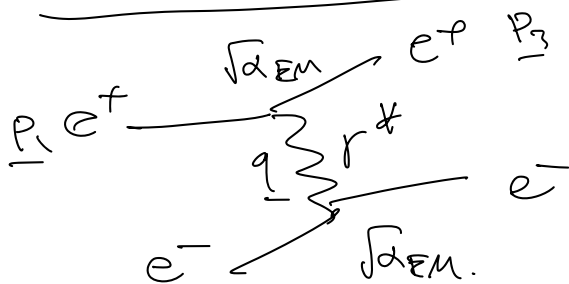
$$NLO \propto (\alpha_s)^2$$

$$NNLO \propto (\alpha_s)^3$$

$$N^3LO \propto (\alpha_s)^4$$



Difference between α_s and α_{EM}



$$M \propto \alpha_{EM}$$

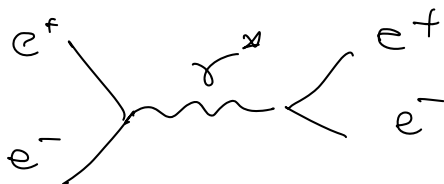
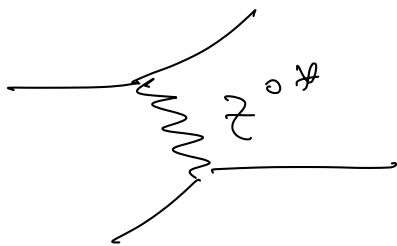
$$\sigma \propto (\alpha_{EM})^2$$

$$q = p_1 - p_3$$

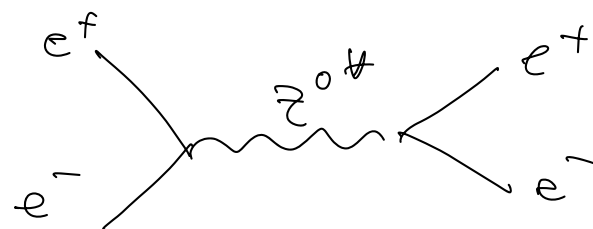
$$q^2 < 0 \quad Q^2 = -q^2 > 0$$

$$Q^2 \approx m_e^2$$

$$Q^2 \approx m_Z^2$$



$$\alpha_{EM}(Q^2 = m_e^2)$$



$$\alpha_{EM}(Q^2 = m_Z^2)$$

$Q^2 \propto$ scale of the process/interaction

$$\alpha_{\text{EM}}(Q^2 = m_e^2) = \frac{1}{137}$$

$$Q^2 = m_e^2 \approx 10^{-7} \text{ GeV}^2$$

$$\alpha_{\text{EM}}(Q^2 = m_Z^2) = \frac{1}{128}$$

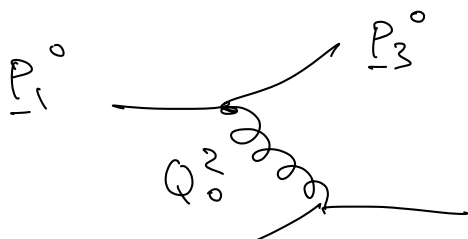
$$Q^2 \approx 10^4 \text{ GeV}^2$$

$\alpha_{\text{EM}} \approx 1\%$ with good precision at all energies.

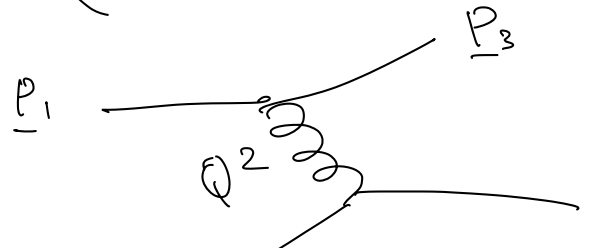
In QCD:

$$\alpha_s(Q^2) \approx \frac{\alpha_s(Q_0^2)}{1 + \frac{7}{4n} \alpha_s(Q_0^2) \log \frac{Q^2}{Q_0^2}}$$

For 3 colors
and 6 flavors of quarks (u, d, s, c, b, t)



$$Q_0^2 = -(p_1 - p_3)^2$$



$$Q^2 = -(p_1 - p_3)^2$$

$$Q^2 \gg Q_0^2$$

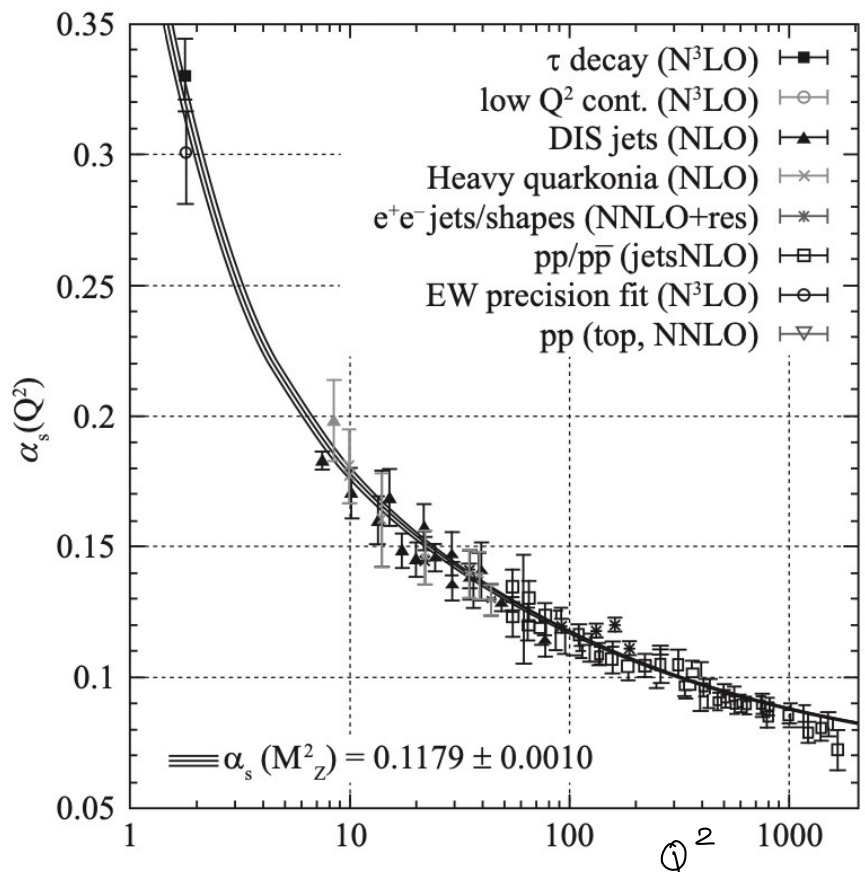


$$\alpha_s(Q^2) < \alpha_s(Q_0^2)$$

Asymptotic freedom
in QCD

$$\alpha_s(M_Z^2) = 0.1179 \pm 0.0010$$

$$\approx \frac{1}{9}$$



Baryon Multiplet

$$B = q_1 q_2 q_3 \quad \psi_B = \psi_{space} \psi_{spin} \psi_{flavor} \psi_{color}$$

A S

L=0

A

hypothesis
Color singlet

$$q_1 \times q_2 \times q_3 = \underbrace{1 \otimes_F}_{\text{antisym under any exchange}} + \underbrace{8 \otimes_{F,12}}_{\text{Sym under flavor exchange } 1 \leftrightarrow 2} + \underbrace{8 \otimes_{F,23}}_{\text{Sym under } 2 \leftrightarrow 3} + \underbrace{1 \otimes_{F,S}}_{\text{Completely Symmetric}}$$

$1 \otimes_{F,S}$

uuu, uud, udu, - -

$\Delta^{++} = uuu$ symm. under flavor exchange.

$$\Delta^+ = uud = \frac{1}{\sqrt{3}} [uud + udu + duu]$$

$$\Lambda = uds = \frac{1}{\sqrt{6}} [uds + sud + dsu + sdu + usd + dus]$$

$$B = q_1 q_2 q_3 \quad S_1 = 1/2 \quad S_2 = 1/2 \quad S_3 = 1/2.$$

$$S = \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \left(\frac{3}{2}\right)_S \oplus \left(\frac{1}{2}\right)_{S,1-2} \oplus \left(\frac{1}{2}\right)_{S,1-3}$$

Symmetric $\uparrow\uparrow\uparrow$
Symm. $1 \leftrightarrow 2$ $\uparrow\uparrow\downarrow$
 $\uparrow\downarrow\uparrow$

$S = \frac{3}{2}$

Symmetric. $\uparrow\uparrow\uparrow$ $\downarrow\downarrow\downarrow$

ψ_{spin} is symmetric.

$$\psi_{F,10}^{S=3/2} = \psi_{space} \psi_{spin} \psi_{flavor} \psi_{color}$$

A S S S A

L=0 ground state

correspond to
10 physical states
observed

$$B = 9_c 9_l 9_3 \quad S_1 = 1/2 \quad S_2 = 1/2 \quad S_3 = 1/2.$$

$$S = \frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \left(\frac{3}{2}\right)_S \oplus \left(\frac{1}{2}\right)_{S,1-2} \oplus \left(\frac{1}{2}\right)_{S,1-3}$$

$\swarrow$ symmetric $\uparrow \uparrow \uparrow$ $\swarrow$ Symm. $\uparrow \leftrightarrow 2$ $\uparrow \uparrow \downarrow$ $\downarrow$ $\uparrow \uparrow \downarrow$

$$S = 1/2$$

$$\psi_{F,10}^{S=1/2} = \psi_{\text{space}} \psi_{\text{spin}}^{S_1,1 \rightarrow 2, 1 \rightarrow 3} \psi_{\text{flav.}} \psi_{\text{color}} \Rightarrow \text{No decuplet with } S=1/2$$

$$A \quad S \quad S_{12} \quad S \quad A$$

$L=0$ S_{12} S_{13} not always symmetric anti-symm under $2 \leftrightarrow 3$

$\Rightarrow$ Decuplet with $S=1/2$ cannot exist

Π_F : Flavor Singlet

$$\psi_B^{\Pi} = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{flav.}} \psi_{\text{color.}}$$

$$A \quad S \quad ? \quad A \quad A$$

$L=0$

$\rightarrow$ Because baryon is a fermion

$\Rightarrow \psi_{\text{spin}}$ must be anti symmetric.

$$S = \left(\frac{3}{2}\right)_S \quad \uparrow \uparrow \uparrow \quad \psi_{\text{spin}} \text{ is symm } \times$$

$$S = \left(\frac{1}{2}\right)_{12} \quad \uparrow \uparrow \downarrow \quad \psi_{\text{spin}} \text{ symm for } 1 \leftrightarrow 2 \quad \times$$

$$S = \left(\frac{1}{2}\right)_{13} \quad \uparrow \downarrow \uparrow \quad \psi_{\text{spin}} \text{ symm for } 1 \leftrightarrow 3 \quad \times$$

$\Rightarrow$ No totally anti-symm. ψ_{spin} .

$\Rightarrow \Pi_F$ cannot exist